

Engineering Report: HDMI-to-Parallel-RGB Board for the T7S150B-E23 (HX8282-A11) 7" Panel using the Lontium LT8619C

TL;DR

- **The LT8619C is not self-configuring; it mandatorily requires an external host/MCU to write its registers over the CSDA/CSCL slave-I2C bus (7-bit address 0x32) at every power-up/reset.** A complete, field-proven register initialization sequence for HDMI-in → 24-bit RGB888 DE-mode out at 1024×600 exists publicly in Hardkernel's GPL u-boot driver (`board/hardkernel/odroid-common/lt8619c.c`), which is your single most valuable resource.
 - **The panel needs five bias rails plus a backlight; a single 5 V input is sufficient and is actually preferable to 12 V** because the best integrated bias IC for this class (TI TPS65150, input range 1.8–6 V) cannot take 12 V. Use TPS65150 for AVDD/VGH/VGL/VCOM and a boost constant-current WLED driver for the backlight (which presents as one ~9.3 V/180 mA load).
 - **No public Lontium register map, programming guide, or reference schematic exists** — Lontium ships these only under NDA. Every register value below is reverse-engineered from open-source driver code; treat them as verified-by-use for 1024×600 RGB888 but request the official *LT8619C Programming Guide* and *Reference Schematic* from `support@lontiumsemi.com` before production.
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Key Findings

1. **Architecture decision (critical):** The LT8619C must be register-initialized by a host every boot. It has an on-chip EDID RAM (written by the host over slave I2C) and a *separate* master-I2C bus (ESDA/ESCL, pins 17/18) labeled "for EEPROM" in the datasheet pin table, but there is no public evidence the chip can auto-boot its full register configuration standalone from that EEPROM. Plan on a small MCU.
2. **I2C slave address = 0x32 (7-bit); 0x64 write / 0x65 read (8-bit).** Your datasheet-guessed 0x92/0x9A is wrong for this control bus.
3. **Register banks are selected by writing register 0xFF:** 0x60 (RX PHY/PLL/CSC/output-timing) and 0x80 (HPD/EDID/clock-detect/audio). **Chip ID:** bank 0x60, reg 0x00=0x16, 0x01=0x04, 0x02≈0xB0 (revision, not validated).
4. **A complete init sequence, a timing-readback register map, and a working on-chip EDID are all published** in the Hardkernel driver — reproduced below.
5. **A single 5 V input powers the whole board.** TPS65150 generates AVDD/VGH/VGL/VCOM with correct internal sequencing; a boost CC driver handles the backlight.
6. **Connectors, HDMI receptacle, ESD arrays, crystal, and MCU** all have concrete, orderable part numbers below.

7. **1024×600 is a non-standard mode**; you must feed the LT8619C's on-chip EDID with a 1024×600 detailed timing and rely on the *source* to output it natively (the LT8619C has no scaler).
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Details

A) LT8619C register-level configuration

Primary source (use this): Hardkernel u-boot, `board/hardkernel/odroid-common/lt8619c.c`, "LT8619C HDMI to RGB888 converter," Copyright 2020 Hardkernel, author CKKim. The header comment states verbatim: *"The base initial code of this driver is from Lontium semiconductor ... output signal of LT8619C is fixed as: resolution 1024x600, 1 port RGB888, 8bit, DE timing mode."* URL: <https://github.com/hardkernel/u-boot/blob/odroidg12-v2015.01/board/hardkernel/odroid-common/lt8619c.c>. A newer Linux V4L2 port exists at `drivers/media/i2c/lt8619c.c` in the `tobetter/linux` `odroid-5.15.y` tree. A second archive (a user-contributed schematic RAR, an N76E003 Keil demo project, and datasheets R1.4/R1.5) is at https://github.com/guitarhua/lt8619C_hdmi_converter.

Documentation status (be explicit): The public LT8619C datasheets (R1.4 17-page, R1.5, R1.7) and the R1.3 Product Brief contain features, pinout, output-mapping tables, and DC characteristics but **no register map, no I2C register descriptions, no init sequence, and no reference schematic**. Everything register-level below therefore comes from the GPL driver code and Odroid boot logs, not from a Lontium document.

I2C addressing. 7-bit **0x32** — confirmed three independent ways: the Odroid device-tree node `lt8619c: hdmi-bridge@32 { compatible="lt8619c"; reg=<0x32>; }`; the driver macro `LT8619C_ADDR`; and working-hardware boot logs printing `i2c master a current slave addr is 0x32`. 8-bit forms = **0x64** (write) / **0x65** (read). Register access is standard 8-bit-subaddress: write `[reg][data]`; read = write `[reg]` then repeated-start read one byte (`aml_i2c_xfer` two-message form in the driver).

Bank/page select. Write register **0xFF** with **0x60** or **0x80** to select the active register page before accessing paged registers. (Seen throughout the driver and echoed by the boot-log line `i2c 0xff 0x60 write failed` when the bus is dead.)

Chip detection.

```
w 0xFF,0x60
r 0x00 -> expect 0x16
r 0x01 -> expect 0x04
r 0x02 -> ~0x80 (revision byte; read but not checked)
```

Healthy hardware prints `Read Chip : 0x16, 0x4, 0xb0`. `ODROID`

Reset/bring-up timing (from driver (`lt8619c_init`)): drive RESET_N (pin 24, active low) low ≥ 5 ms (`udeelay(5000)`), then high, then wait ~ 10 ms (`udeelay(10000)`) before the first I2C access. Reading the ID before this delay is the documented cause of all-zero ID reads.

Full initialization sequence for HDMI $\rightarrow$ RGB888, 3.3 V TTL, DE mode (reproduced from the driver; all writes are reg.value in the currently selected bank):

Power-on / EDID / HPD phase (`lt8619c_poweron`):

```
// HPD low while loading EDID (lt8619c_setHPD(0))
0xFF,0x80 ; 0x06 &= ~0x08      // RX_HPD low
// load 256-byte on-chip EDID (lt8619c_set_edid)
0xFF,0x80 ; 0x8E,0x07 ; 0x8F,0x00
0x90 <- 256 EDID bytes (block write)
0x8E,0x02
// (delay 100 ms)
0xFF,0x80 ; 0x06 |= 0x08      // RX_HPD high -> source now reads EDID
```

RX PHY / PLL / clock init (`lt8619c_rx_init`):

```
0xFF,0x80 ; 0x2C |= 0x30      // RGD_CLK_STABLE_OPT[1:0]
0xFF,0x60
0x04,0xF2
0x83,0x3F
0x80,0x08                    // use xtal_clk (25 MHz) as sys_clk
0xA4,0x10                    // 0x10 = SDR pixel-clock output; 0x14 = DDR
0x07,0xFF
0xA8,0x0F
0x60,0x00
0x96,0x71
0xA0,0x50
0xA3,0x74                    // phase-adjust enable + route PCLK to output pin
0xA2,0x29                    // PCLK phase code (choices:
0x20,0x28,0x21,0x29,0x22,0x2a,0x23,0x2b,0x24,0x2c)
0x6D,0x00                    // RGB output mapping (0x00 = default D0-7=R, D8-
15=G, D16-23=B)
0x6E,0x00                    // RGB high/low bit-swap control
// enable RGB888 output FIFO
0x0E,0xFD ; 0x0E,0xFF ; 0x0D,0xFC ; 0x0D,0xFF
```

RX reset (`lt8619c_rx_reset`), run if PLL not locked):

```

0xFF,0x60
0x0E,0xBF // reset RXPLL
0x09,0xFD // reset RXPLL lock-det (delay 5 ms)
0x0E,0xFF ; 0x09,0xFF
0x0E,0xC7 // reset PI
0x09,0x0F // reset RX, CDR (delay 10 ms)
0x0E,0xFF (delay 10 ms)
0x09,0x8F (delay 10 ms)
0x09,0xFF (delay 50 ms)

```

Clock / PLL status (poll after reset):

```

0xFF,0x80 ; r 0x44 & 0x08 -> RX clock stable
0xFF,0x80 ; r 0x87 & 0x10 -> RXPLL locked

```

Input-info readback (HDMI infoframes; `lt8619c_input_info`, bank 0x80):

```

r 0x13 bit0 = Hsync stable ; bit1 = HDMI(1)/DVI(0) mode
r 0x74 & 0x7F = input VIC
r 0x71 & 0x60 = colorspace (RGB / YCbCr444 / 422)
r 0x16 & 0xF0 = color depth
r 0x72 & 0xC0 = colorimetry ; r 0x73 = quant range / ext colorimetry
r 0x75 & 0x0F = pixel-repeat factor

```

Color-space conversion (`lt8619c_csc_conversion`, bank 0x60): write 0x07,0xFE then program 0x52/0x53 per input colorspace & quant range (full decision table in source). For RGB-in → RGB-out full-range: 0x52,0x00 ; 0x53,0x00.

Detected input timing readback (`lt8619c_check_video`, bank 0x60):

Parameter	Registers
H active	0x22 (MSB), 0x23 (LSB)
V active	0x20[3:0], 0x21
H sync width	0x14[3:0], 0x15
V sync width	0x13
H back porch	0x18[3:0], 0x19
V back porch	0x16
H total	0x1E, 0x1F
V total	0x1C[3:0], 0x1D
Frame counter	0x10, 0x11, 0x12
Sync polarity	0x24 bit0 = Hsync, bit1 = Vsync
TMDS clock (kHz)	bank 0x80: 0x44[2:0], 0x45, 0x46

Output timing programming (`(lt8619c_bt_setting)`, bank 0x60): reg 0x60 sets output H/V sync polarity (|0x20 Hsync pos, |0x10 Vsync pos) and interlace doubling; registers 0x61–0x66 program output H (sync+backporch, active, total) and 0x67–0x6C program output V. In this reference the output timing is copied from the detected input timing, giving genlocked pass-through.

Audio (optional, `(lt8619c_audio_init)`): 0xFF,0x60;0x4C,0x00 (I2S 2-ch) or 0x80 (SPDIF); then 0xFF,0x80;0x5D,0xC9;0x07,0x16;0x08,0x80.

3.3 V vs 1.8 V TTL output level is a hardware choice: power the `(vcc33_ttl)` rail at 3.3 V for 3.3 V logic (matches the panel's DVDD 3.3 V, so no level shifting on the 24-bit bus). **Output driver strength/slew** registers are not in any public document. **Disabling LVDS when using TTL**: the reference RGB path simply never enables the LVDS transmitter (`(detect_lvdspll_lock)`/`(lt8619c_bt_setting)` gate on `(0x60 A3 bit6)`); there is no published explicit "LVDS off" bit. **RGB888 default mapping** (D0–7=R, D8–15=G, D16–23=B) is the reset default, kept by 0x6D,0x00 / 0x6E,0x00.

HDCP: the LT8619C decrypts HDCP 1.4 automatically once configured; keys are on-chip, so no HDCP-key EEPROM is used in the reference design. No public register controls HDCP enable in the open driver.

Interrupt / hot-plug: RX_HPDP is driven by firmware via bank 0x80 reg 0x06 bit3; +5 V cable detect is a board-level input to the MCU (see hardware section). Clock-appear/disappear is polled

via bank 0x80 reg 0x44 bit3; PLL lock via 0x87 bit4.

Self-boot vs MCU (explicit evidence): The datasheet R1.5 pin table lists pin 17 ESDA / pin 18 ESCL as "Master I2C data/clock channel for EEPROM, 5V-tolerant," and pin 65 CSDA / pin 66 CSCL as "Slave I2C data/clock channel." The Features list includes "Support On-chip EDID." In the reference design the host writes the EDID into on-chip EDID RAM (reg 0x90) and pushes the entire register configuration over the slave bus at every boot — `lt8619c_init()` resets the chip, checks the ID, then runs power-on/EDID/rx-init/timing on every power cycle. **There is no code path or datasheet text describing standalone auto-configuration from the EEPROM.**

Conclusion: budget for an MCU. If you populate a 24Cxx on ESDA/ESCL at all, treat it (per general EDID convention) as an EDID store at 7-bit 0x50 / 8-bit 0xA0 — but this is unconfirmed for the LT8619C and unnecessary given the on-chip EDID. [elecfans](#) [Lontiumsemi](#)

Known gotchas (Odroid / ArchLinuxARM forums & kernel list):

- First bank-select write (0xFF←0x60) times out (`ret=-110`) or NAKs (`ret=-5`) and chip ID reads `0x0,0x0,0x0` → "failed to read LT8619C chip id." Root causes: reset/power sequencing, or reading ID before the ~10 ms post-reset delay. (forum.odroid.com t=44365; [pastebin.com/NQq2aeLH](#)) [Pastebin](#)
- Bus contention: on shared buses a powered-down downstream master (e.g., a Pi CM5 with 1.8 kΩ pull-ups on the DDC pins) can hold the bus and block MCU→LT8619C traffic. Give the LT8619C a private I2C segment. (raspberrypi forums t=381487)
- The mainline candidate driver has a cosmetic bug (`((pbuf+127)==NULL)` always false) flagged by the kernel test robot — harmless. (lore.kernel.org 202402251842) [Linux Kernel](#)

B) LT8619C reference schematic / hardware

No official reference schematic is public. A user-contributed schematic is in [LT8619C设计资料.rar](#) at the guitarhua repo (unverified). Guidance below combines the datasheet pin/DC data with general HDMI-RX practice — flagged as inference where noted.

Power rails & decoupling. The LT8619C has 1.8 V rails (VCCA18, VDD18, PVCC18) and 3.3 V rails (VCCA33, VCC33, VCCA33_XTAL, VCC33_TTL). Standard practice (inference — confirm against the Lontium reference): one bulk 10 μF per voltage domain plus a 100 nF (and a 10 nF for the analog/PLL rails) immediately at each pin; keep VCCA33_XTAL and PVCC18 (PLL) especially quiet with a ferrite bead + local 100 nF + 1 nF. Tie the exposed pad to a solid ground pour with a via array. [elecfans](#)

Power sequencing (DIFF3318 constraint). The datasheet requires any 3.3 V pin minus any 1.8 V pin to stay within -1.0 V to +2.0 V, and same-voltage pins within 1.0 V (DIFF33/DIFF18). The safe approach is to bring the 1.8 V and 3.3 V rails up close together: generate 1.8 V from the 3.3 V rail with an LDO whose enable makes 1.8 V follow 3.3 V within a few hundred μs, or use a dual-output PMIC/tracking regulator. Avoid letting 3.3 V stand fully up with 1.8 V at 0 V for an extended interval. (Exact sequence/timing is in the NDA reference design — request it.)

VTERM (pins 4 & 10) tie to 3.3 V as the TMDS termination supply. **REXT** (pin 16) needs 2.0 kΩ 1% to ground. **RESET_N** (pin 24) gets an RC (e.g., 10 kΩ to 3.3 V + 100 nF) plus an MCU GPIO override for the driver's 5 ms/10 ms sequence. (elecfans)

HDMI front end. AC-couple each TMDS pair with 100 nF (0402) series caps near the connector; keep 100 Ω differential; place the ESD array right at the connector. HPD (RX_HPD) is an *output from the LT8619C toward the source* — level-shift/pull it to the incoming HDMI +5 V through a small MOSFET or dedicated HPD buffer. DDC (DSDA_RX/DSCL_RX, pins 75/76): 2.0 kΩ pull-ups to 3.3 V. +5 V detect: divide the HDMI +5 V (e.g., 100 k/47 k) into an MCU GPIO. CEC (if used): 27 kΩ pull-up.

25 MHz crystal (XTAL_I/XTAL_O, pins 61/60). Fundamental-mode 25 MHz, load capacitance $CL \approx 8$ pF, tolerance ± 30 ppm (± 50 ppm acceptable), ESR < 50 Ω. Load caps per pin $\approx 2 \times (CL - C_{stray})$; with ~ 3 pF stray, ~ 10 – 12 pF each — tune on the bench (parallel-resonant load-cap matching directly sets frequency accuracy). Keep traces short and guard with ground.

Thermal / current. QFN76 with exposed pad; consumption is dominated by the TMDS receiver and PLL. Public docs give no single "typical current" figure — request it from Lontium; budget a few hundred mW and provide the exposed-pad thermal via array. (elecfans)

C) EDID for a 1024×600 @ 60 Hz sink

Because the LT8619C has no scaler, the **source** must output 1024×600 natively; the EDID you place in on-chip EDID RAM is what convinces the source to do so.

Recommended native timing (true 60.0 Hz): pixel clock **51.20 MHz**, H: active 1024, blank 320, total 1344; V: active 600, blank 35, total 635 $\rightarrow 51.20\text{M}/(1344 \times 635) = 59.98$ Hz. This sits comfortably inside the panel's DE-mode ranges (DCLK 40.8–67.2 MHz, H total 1114–1400, V total 610–800). Real-world confirmation: an existing 1024×600 module's EDID decodes to exactly "Clock 51.210 MHz ... 1024 1056 1072 1344 ... 600 616 619 635 ... +hsync -vsync, VertFreq 60 Hz" (Odroid forum t=44016). Hardkernel's shipped table instead uses 50.40 MHz / $1344 \times 638 = 58.8$ Hz, which also works; the 51.20 MHz/635 variant is cleaner for 60 Hz. (GitHub)

Detailed Timing Descriptor (18 bytes) computed for 51.20 MHz, hfp 32 / hs 16, vfp 13 / vs 6, image 154×86 mm, both syncs negative (DE mode makes polarity a don't-care at the panel):

00 14 00 40 41 58 23 20 20 10 D6 00 9A 56 00 00 00 18

Byte meaning: **00 14** = pixel clock 5120 ($\times 10$ kHz); **00** = Hactive LSB; **40** = Hblank LSB (320); **41** = Hactive/Hblank MSN (1024/320); **58** = Vactive LSB (600); **23** = Vblank LSB (35); **20** = Vactive/Vblank MSN; **20** = Hfront (32); **10** = Hsync (16); **D6** = Vfront(13)/Vsync(6) low nibbles; **00** = high bits; **9A/56** = 154/86 mm; **00** = size MSN; **00 00** = borders; **18** = digital separate sync, H– V–. (This mirrors the field layout documented in VESA E-EDID and confirmed by the NXP EDID walkthrough.)

Checksum: byte 127 = $(256 - (\text{sum of bytes } 0..126 \bmod 256)) \bmod 256$, so the whole 128-byte block sums to 0 mod 256. The Hardkernel driver computes both the DTD and the checksum programmatically (`lt8619c_edid_calc`) writes the DTD at offset 0x36; `lt8619c_edid_checksum` fills byte 127) — reuse that code rather than hand-typing.

Reference on-chip EDID. The driver ships a full 256-byte EDID (`onchip_edid[]`) whose base block advertises a 1024×768 fallback and whose first DTD is overwritten at runtime with the 1024×600 timing at offset 0x36, plus a CEA-861 extension block. That array is a ready-made template. [GitHub](#)

Extension block / CEA-861. For a pure "monitor" that needs only one mode, a single 128-byte EDID 1.3 block with the 1024×600 DTD as the *preferred* (first) timing is the most compatible. Add a CEA-861 extension only if you need HDMI audio or want to flag basic audio; a VSDB is not required for video. Trade-offs: PC GPUs (Windows/Linux) and the Raspberry Pi honor the preferred DTD well; some Android phones and cheap media players ignore non-CEA DTDs and only emit CEA short-video-descriptor modes (720p/1080p) — those sources will not produce 1024×600 and are fundamentally incompatible without a scaler. Also note the historic driver limitation that some GPUs only use *standard timing* descriptors whose resolution must be divisible by 8 (1024 and 600 both qualify, so you are safe here).

Fallback timings to also list (Standard/Established Timings, since the panel tolerates 40.8–67.2 MHz, H 1114–1400, V 610–800): 1024×768@60 (65 MHz — near the top of the DCLK range but valid), 800×600@60, 640×480@60. These will be letterboxed/mis-scaled unless the panel is fed exact pixels; they exist mainly to keep stubborn sources from falling back to no output.

Tools: `edid-decode` (validate/encode; Linux), AW EDID Editor (Windows GUI), wxEDID, Deltacast E-EDID Editor, and `linuxhw/edid-generator` or `akatrevorjay/edid-generator` (Makefile-driven, takes a modeline; the latter was used by an Odroid user to build a 1024×600 binary from decoded values). Always verify the binary with `edid-decode` before flashing.

D) TFT bias supply (AVDD 9.6 V, VGH 18 V, VGL –6 V, VCOM 3.1 V) from 5 V

5 V vs 12 V — use 5 V. The best-matched integrated bias IC, TI **TPS65150**, per its datasheet (SLVS576, Rev. B) "*With an input voltage range of 1.8 V to 6 V the device is ideal for ... monitor applications with a 5-V input voltage rail*" — a 12 V input would exceed its rating and force a lesser part. 12 V's only benefits (backlight becomes a step-down; lower input current) are minor for a ~2–3 W panel and not worth a second high-voltage rail. Keep the single 5 V input.

Recommended bias IC: TI TPS65150 / TPS65150-Q1 (HTSSOP-24 / VQFN, integrated VCOM buffer, 2-A internal boost MOSFET producing the main output V(VS) up to 15 V at <1% accuracy, plus two regulated adjustable charge pumps for VGH and VGL, with adjustable VGH/VGL power-on sequencing built in). It is purpose-built for this exact rail set on 7"–10" panels. The panel's VGH +18 V / VGL –6 V at ~1–5 mA sit well within the part's demonstrated capability (EVM guide SLVUAG5A characterizes "VGH = 24 V, Iout 0–20 mA" and "VGL = –5 V, Iout 0–20 mA").

- **Alternatives:** TPS65105 (input 2.7–5.8 V, integrated VCOM buffer, 1.6-MHz PWM boost, 220-nF charge-pump caps; datasheet: *"The TPS65100/01 has a typical switch current limit of 2.3 A, and the TPS65105 has a typical switch current limit of 1.37 A"* — good, but fixed internal sequencing that may not meet a >10 ms VGH-after-VGL delay, per TI E2E thread on this exact question). TPS65100/TPS65101 (2.3-A boost, VCOM buffer). TPS65130 (dual $\pm$ output, no VGH/VGL charge pumps — less ideal). Richtek RT9911/RT9910 and MPS MP1541-class are functional China-market equivalents; common low-cost drop-ins on 7" RGB modules include **Fitipower FP6717/FP6718** and **SGMICRO SGM6130-class**. Verify 2026 stock at LCSC/Digi-Key before committing.
- **Component values (TPS65150 typical app):** boost inductor 3.3–6.8 μ H (e.g., 4.7 μ H, $I_{sat} \geq 1$ A); main + charge-pump Schottkys ~ 40 V (SS14 for the boost, BAT54-class for the low-current pumps); charge-pump flying/reservoir caps 220 nF–1 μ F X7R rated ≥ 25 V (VGH) and $\geq \sim 16$ V (VGL); boost output cap 22 μ F ceramic (TI's recommended value); feedback dividers set each rail per the datasheet equations. Because VGH/VGL/VCOM draw little current, keep ceramic caps high-voltage-rated to fight DC-bias derating and overshoot.

VCOM = 3.1 V $\pm$ 0.1 V, adjustable to minimize flicker. The TPS65150's on-chip VCOM buffer output is set by a resistor divider into its VCOM feedback; for production flicker trimming use a **digital VCOM calibrator** (e.g., a Renesas/Intersil ISL45041/ISL45042 or an ADI/Maxim VCOM DAC) driving the buffer input so VCOM is one-time-programmable at end-of-line. A trimpot is fine for prototypes but not production-friendly. The panel exposes VCOM on FPC pins 6 and 46 and DITHB on pin 47 — drive VCOM from the buffer and set DITHB per the panel's dithering requirement.

Power sequencing for HX8282-A11-class panels (cite Himax HX8282-A / -A02 / -A14 datasheets — the -A11 is the same 1536-channel source-driver family). The Himax driver has a built-in power on/off sequence controller and, per the HX8282-A datasheet, *"has a power on/off sequence control function. In order to prevent IC from power on reset fail, the rising time (TPOR) of the digital power supply VDD should be maintained within the given specifications."* Recommended order **UP:** DVDD (3.3 V) $\rightarrow$ AVDD (9.6 V) $\rightarrow$ VGL (–6 V) $\rightarrow$ VGH (+18 V) $\rightarrow$ VCOM $\rightarrow$ release RSTB (RC 10 k Ω /1 μ F, min 50 μ s pulse) $\rightarrow$ apply valid DCLK/DE/data $\rightarrow$ **backlight LAST. DOWN:** reverse — backlight off $\rightarrow$ stop video $\rightarrow$ assert RSTB $\rightarrow$ VCOM $\rightarrow$ VGH $\rightarrow$ VGL $\rightarrow$ AVDD $\rightarrow$ DVDD. The TPS65150's VGL-before-VGH behavior matches this. Exact inter-rail delays (T1–T6) and TPOR for the -A11 are in the Himax HX8282-A11 datasheet (DOC No. HX8282-A11-DS) — obtain it for precise values.

- **Per-rail current (7" 1024 $\times$ 600, estimate for sizing):** DVDD 3.3 V $\sim$ 30–80 mA; AVDD 9.6 V $\sim$ 20–50 mA; VGH +18 V $\sim$ 1–5 mA; VGL –6 V $\sim$ 1–5 mA; VCOM a few mA transient. The TPS65150 handles all of these with margin.

E) Backlight LED driver

Array topology note. VLED typ 9.3 V with ILED 180 mA total and a *single* A/K pair is consistent with **3 series $\times$ 9 parallel** (9.3 V $\approx$ 3 $\times$ 3.1 V; 180 mA $\approx$ 9 $\times$ 20 mA), **not** 9-in-series (which would be

~28 V). Because only one A/K pair is exposed, the internal strings are paralleled and you drive the whole 27-LED array as **one ~9.3 V / 180 mA constant-current load** — there is no alternative, and current sharing among the internal parallel strings relies on the panel's Vf matching.

Recommended driver: Diodes AP3031 (SOT-23-6, 1 MHz). Per its datasheet (DS41442 Rev.2): *"1.4A switching current allows AP3031 to be used in different 7' to 10' LCD panel backlights (3×13 LED arrays typically) ... 1 mm tall 4.7 μH inductor and 0.47 μF output capacitor for the typical application is sufficient,"* with a wide 2.7–16 V input and a *"Low 200 mV Feedback Voltage."* So $R_{SENSE} = 0.200\text{ V} / 0.180\text{ A} \approx 1.11\ \Omega$ (1 Ω gives ~200 mA; pick 1.1 Ω 1% for 182 mA), cycle-by-cycle limit 1.4 A. Pricing is ~\$0.12 (AP3031KTR-G1). Use a 4.7 μH / ≥0.7 A inductor, a ≥20 V Schottky if not integrated, and a 16 V-rated output cap. Alternatives: MPS MP3202/MP3208-class, SGMICRO SGM3735, or a generic boost-CC.

- **PWM dimming.** Drive the driver's EN/CTRL (or a series dimming FET) with an MCU PWM. Choose ~**2–5 kHz** — above the audible band (>2 kHz avoids ceramic-cap/inductor "singing") yet low enough for the boost to recover each cycle; avoid the 200 Hz–2 kHz audible region and avoid frequencies that beat with the 60 Hz frame rate to prevent visible flicker. If audible noise or EMI persists, raise to ~10–20 kHz. **Do not** use the Diodes AL3050 here — it is a U-DFN2020 part limited to ~40 mA/string and is unsuitable at 180 mA (noted so it isn't mistakenly selected).

F) Connectors, HDMI receptacle, ESD

50-pin 0.5 mm FPC connector (panel FPC 0.3 ±0.03 mm thick, ~25.5 mm wide):

- **Hirose FH12S-50S-0.5SH(55)** — 50-pos, 0.5 mm, right-angle SMT, ZIF flip-lock, **bottom contact**, reinforced actuator (the reinforced "S" version is recommended for 50-pin to resist actuator breakage). RS 738-8843P. Standard (non-reinforced): **FH12-50S-0.5SH(55)**. Vertical variant: **FH12-50S-0.5SVA(54)**. FFC/FCB thickness spec 0.30 mm matches your FPC. ([Rsdelivers](#))
- **Top-contact** equivalent: **Hirose FH12A** series. ([HIROSE Electric](#))
- **Contact-side rule:** with a **bottom-contact** connector, the FPC's exposed contacts must face **down** (toward the PCB) at insertion; with a top-contact connector they face up. Confirm the T7S150B-E23 FPC contact face from its mechanical drawing and choose accordingly, or the 24-bit bus will be mirrored.
- **Lower-cost equivalents:** JAE FI-X, Molex 5054xx / 200528, Amphenol; Chinese: **Jushuo AFC01-S50FCA-00**, XKB, HDGC (all widely stocked at LCSC/JLCPCB for economy assembly).

HDMI Type-A receptacle: SMT with through-hole shell pegs — **Amphenol 10029449-001RLF**, **Molex 47151-0001**, or **GCT HDMI001** family; through-hole versions exist in the same families. Pick one with strong shield-tab retention.

TMDS ESD: place a low-capacitance TVS array right at the connector. **TI TPD4E05U06** (or **ESD204** pairs — TI's HDMI app note SBOA280 uses two ESD204 + one TPD4E05U06 for a full HDMI 2.0 port), or **Semtech RClamp0574P** (one device protects two pairs, so two devices cover the 3 data + 1 clock pairs). ST offers HDMI-specific arrays too. Keep $< \sim 0.5$ pF/line and 100 Ω differential through the protection, and route the protection as a flow-through stub-free layout.

G) MCU choice

You need a small I2C master (100/400 kHz), a few GPIOs (RESET_N, HPD helper, +5 V detect, backlight/bias enables) and one timer PWM for dimming.

- **Best fit / your STM32 preference: STM32C011F6P6** (TSSOP-20, Cortex-M0+; the STM32C0 line is ST's cheapest, from $\sim \$0.21$ and $\$1.02/500$ -piece on ST's eStore) or **STM32G031J6M6** (SO-8, if you want minimal pin count). The two are pin-map-consistent. **STM32C031/STM32F030** if you want more flash/RAM. Free toolchain: **STM32CubeIDE** (GCC), or **libopenm3**. ([GroupGets](#))
 - **Ultra-low-cost alternatives: PY32F002/PY32F003** (Cortex-M0+, pennies), **CH32V003** (RISC-V, $\sim \$0.10$ – 0.15 , free GCC via MounRiver / [ch32v003fun](#)), or **STC8** (8051). For a Western low-volume supply chain, STM32C011 or CH32V003.
 - **Reusable firmware:** the Hardkernel u-boot ([lt8619c.c](#))/([lt8619c.h](#)) (GPL) is directly adaptable — port ([lt8619c_reg_read/write](#)), ([lt8619c_check_chipid](#)), ([lt8619c_poweron](#)), ([lt8619c_rx_init](#)), ([lt8619c_detect_clk](#)), ([lt8619c_check_video](#)), ([lt8619c_bt_setting](#)) onto STM32 HAL/LL I2C. The Linux ([drivers/media/i2c/lt8619c.c](#)) and the guitarhua N76E003 Keil demo are secondary references. No dedicated open-source STM32 LT8619C project is known; you will adapt the u-boot code.
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Recommendations (staged)

Stage 1 — De-risk the bridge (do this first). Build/buy a minimal LT8619C board (HDMI-in, LT8619C, 25 MHz xtal, 3.3 V/1.8 V rails, RGB on a header), wire an STM32C011 on a *private* I2C bus, and port the Hardkernel init. **Success criteria:** chip ID reads ([0x16, 0x04, 0x??](#)); bank 0x80 reg 0x44 bit3 shows clock-stable and reg 0x87 bit4 shows PLL-locked when a source outputs your 1024×600 EDID; recovered TMDS clock (bank 0x80 regs 0x44–0x46) ≈ 51.2 MHz. If a source won't emit 1024×600, that's an EDID/source-compatibility issue, not a board bug — test with a Linux PC first (most tolerant of custom DTDs).

Stage 2 — Panel bias + power sequencing. Bring up TPS65150 (AVDD/VGH/VGL/VCOM) on the bench with the panel disconnected; verify rails and sequencing order. Set VCOM to 3.1 V and trim for minimum flicker once video is live. Only then connect the panel FPC.

Stage 3 — Backlight + integration. Add the AP3031 at ~ 182 mA (R_{SENSE} 1.1 Ω), PWM at ~ 2 – 5 kHz, backlight enabled **last**. Integrate MCU control of the full power-up/down order.

Stage 4 — Production hardening. Add digital VCOM calibration, finalize EDID (validate with `edid-decode`), and **before committing the layout/BOM, email support@lontiumsemi.com for the official LT8619C Programming Guide, Reference Schematic, and typical-current/thermal data** to confirm the power-sequencing (DIFF3318), decoupling, and any output-slew registers.

Thresholds that change the plan: if your target sources are phones/media players that only emit CEA modes, the LT8619C (no scaler) is the wrong chip — switch to a *scaling* bridge (RTD2556/RTD2668-class, or a Lontium scaler part such as LT8619B which integrates a scaler + MCU). If you are forced to a 12 V input, drop TPS65150 for a higher-Vin bias IC and make the backlight a buck.

Caveats

- **Every LT8619C register value here is reverse-engineered from GPL driver code, not an official register map.** It is proven for 1024×600 RGB888 DE-mode but is not guaranteed complete — output slew, an explicit LVDS-disable bit, and HDCP-enable bits are absent from all public sources. Obtain the NDA *LT8619C Programming Guide* to fill these gaps. Do not assume undocumented registers.
- **No official LT8619C reference schematic is public.** Power-sequencing (DIFF3318), decoupling specifics, HPD level-shift, and typical current are inference or general practice — confirm against the Lontium reference design.
- The "**3 strings × 9 series**" wording in your panel notes is inconsistent with VLED 9.3 V; treat the backlight as a single 9.3 V/180 mA node (likely 3-series × 9-parallel internally). Confirm from the panel drawing before finalizing the LED driver's OVP setting.
- The **HX8282-A11 datasheet is not fully public**; sequencing/current figures are drawn from sibling parts (HX8282-A/-A02/-A14). Get the -A11 datasheet for exact T1-T6 timing and TPOR.
- Bias-IC and LED-driver **2026 availability/pricing** should be re-checked at LCSC/Digi-Key/Mouser; several MPS backlight controllers (MP3388, MP3394) are marked "not recommended for new designs."
- **Source URLs for verification:** Hardkernel driver — <https://github.com/hardkernel/u-boot/blob/odroidg12-v2015.01/board/hardkernel/odroid-common/lt8619c.c>; guitarhua archive — https://github.com/guitarhua/lt8619C_hdmi_converter; LT8619C R1.5 datasheet — <https://file.elecfans.com/web2/M00/87/6D/poYBAGOuTB2AHZ7fAArRqvmLhoo942.pdf>; Product Brief R1.3 — https://www.lontiumsemi.com/UploadFiles/2022-07/LT8619C_Brief_R1.3.pdf; TPS65150 — <https://www.ti.com/product/TPS65150>; TPS65105 — <https://www.ti.com/product/TPS65105>; AP3031 — <https://www.diodes.com/part/view/AP3031>; HX8282-A datasheet — https://www.phoenixdisplay.com/wp-content/uploads/2019/04/HX8282-A_DS_Preliminary_v02.pdf; HX8282-A11 — <https://www.seacomp.com/sites/default/files/references/Himax-HX8282-A11.pdf>; TI HDMI ESD app note SBOA280 — <https://www.ti.com/lit/pdf/sboa280>; Hirose FH12 —

<https://www.hirose.com/en/product/series/FH12>; 1024×600 EDID reference decode —
<https://forum.odroid.com/viewtopic.php?t=44016>; VESA E-EDID standard —
<https://glenwing.github.io/docs/VESA-EEDID-A2.pdf>.